

BanglaSentNet: A Hybrid Deep Learning Framework for Multi-Aspect Sentiment Analysis in Bangla E-Commerce Reviews

Ariful Islam^{1*} and Md Rifat Hossen²

^{1*}Computer Science and Engineering, Chittagong University of Engineering and Technology, Chittagong, Bangladesh.

²Computer Science and Engineering, Chittagong University of Engineering and Technology, Chittagong, Bangladesh.

*Corresponding author(s). E-mail(s): arifulislamnayem11@gmail.com;
Contributing authors: rifat8851@gmail.com;

Abstract

Product reviews in Bangla text pose major analytical challenges due to the insufficient availability of language resources and benchmarked datasets. This research presents an upgraded multi-label category-based sentiment analysis framework for Bangla based on a hybrid deep learning approach. Our methodology combines multiple deep learning architectures, including LSTM, BiLSTM, and GRU, along with GloVe and FastText word embeddings and the transformer model BanglaBERT. Our proposed weighted ensemble strategy effectively balances model predictions, resulting in improved classification accuracy. The framework achieves a weighted F1 score of 0.88, outperforming standard machine learning models such as Logistic Regression (LR), Support Vector Machine (SVM), Naive Bayes (NB), AdaBoost, Gradient Boosting (GB), and Random Forest (RF), while also exceeding individual deep learning baselines including CNN, LSTM, BiLSTM, GRU, and CNN+BiLSTM. This research establishes a strong solution for sentiment analysis in Bangla through scalable e-commerce customer feedback analysis that improves user satisfaction and enables better decision-making.

Keywords: Sentiment analysis, Multi-label text classification, Deep learning, Ensemble learning, Bangla language, Product reviews, Transformer models

1 INTRODUCTION

In today’s digital marketplace, online product reviews have fundamentally transformed consumer behavior and purchasing decisions. Product analysis on the basis of customers’ opinions has become a must to make the right decision by obtaining product quality, price, and service information. Customer opinions assist companies with strategic decision-making as well as in enhancing customer relationship management. Though a great amount of research has been conducted in sentiment analysis in most of the languages, category-based sentiment classification for Bangla is not thoroughly explored yet. Bangla is spoken by more than 300 million people globally and is struggling with a massive digital divide in terms of natural language processing capabilities. Most traditional sentiment analysis models work mainly with general polarity judgments and do not reflect the specific sentiments found in separate review categories. Real-world product reviews tend to be complex, using various facets and contradictory thoughts within one piece of writing. Consider the following Bangla product review:

Bengali: পণ্যের মান বেশ ভালো, তবে দাম অনেক বেশি। ডেলিভারি সার্ভিস চমৎকার ছিল কিন্তু প্যাকেজিং একটু দুর্বল।

English: The product quality is quite good, but the price is too high. The delivery service was excellent, but the packaging was somewhat weak.

This opinion carries both positive sentiment regarding the quality of products and delivery service and negative sentiment regarding price and packaging. Such sophisticated sentiment expressions should be identified by models capable of making accurate multidimensional judgments across product domains. Bangla sentiment analysis is confronted with several challenges. The scarcity of fully annotated datasets, standard preprocessing tools, and domain-specific language resources is a key hindrance. Bangla text is rich morphologically, has colloquial writing behaviors, frequent code-mixing with English, and limited pre-trained language models specifically fine-tuned for Bangla processing. Deep learning methods and transformer-based models have shown remarkable performance in NLP tasks in recent years. Most developments have been for well-resourced languages, and under-resourced languages like Bangla are faced with very limited access to state-of-the-art methods. The introduction of language-specific transformers like BanglaBERT opened new horizons, but their application in domain-specific tasks remains to be unexplored. His paper introduces BanglaSentNet, a hybrid deep learning network for multi-aspect sentiment analysis of Bangla e-commerce reviews. Our approach employs a multilabel classification paradigm that identifies sentiments in four key categories: quality, service, pricing, and decoration. The method employs an ensemble of LSTM, BiLSTM, GRU, and BanglaBERT models with a weighted ensemble approach using different word embeddings like GloVe, Fast-Text, and transformer-based representation. The key contributions of this research include:

- Development of a large-scale dataset comprising Bangla product reviews with sentiment labels organized by categories.
- Design of a deep learning model that integrates LSTM, BiLSTM, GRU, and BanglaBERT to improve classification accuracy.
- Implementation of a weighted ensemble approach that unites various models to enhance sentiment classification quality.
- Solutions addressing Bangla NLP challenges through preprocessing and feature engineering to handle class imbalance, misspellings, informal language, and limited labeled data.

This research complements Bangla NLP capability while providing practical utilities for market research analysts and e-commerce websites to better leverage customer feedback analysis insights. This research advances the democratization of NLP tools for under-resourced languages in commercially relevant use cases.

2 Related Work

Sentiment analysis has advanced considerably with the advent of deep learning techniques and aspect-based approaches. Bangla language multi-aspect sentiment analysis in general, and particularly in the e-commerce sector, however, remains unexplored. This section overviews the research status in sentiment analysis globally and from the Bangla viewpoint, presenting notable methodology contributions and areas still to be explored.

2.1 Aspect-Based Sentiment Analysis: International Foundations

Standardization of assessment frameworks for aspect-based sentiment analysis (ABSA) has predominantly been done by the SemEval workshops. Pontiki et al. [1] initiated Task 12 for aspect-based sentiment analysis, developing large benchmarks for sentiment classification at the aspect level. This effort was further pursued by Pontiki et al. [2] in Task 5, introducing standardized datasets that enabled systematic comparison across differing methodological frameworks. Learning on such fundamentals, Tsai et al. [3] constructed a sophisticated system for SemEval-2014 Task 4, aspect-opinion relation analysis. Their approach showed the necessity of identifying relations between aspects and corresponding sentiment polarities, establishing baseline methods for category-based sentiment detection. Employing graph-based techniques marked a highlight for ABSA research. Cai et al. [4] proposed Hier-GCN, a graph convolutional network that is capable of effectively extracting explicit and implicit sentiment aspects from textual content. Their work proved the ability of graph neural networks to model complex sentiment relationships and deliver substantial performance gain on benchmark SemEval datasets. Domain-specific use cases have been the recent scope of contributions. Maroof et al. [5] suggested a hybrid approach combining lexical and machine learning techniques to perform aspect-based sentiment analysis in the service industry. Their solution was found to provide higher quality assessments for customer feedback analysis, highlighting real-world business usage of ABSA techniques.

2.2 Bangla Natural Language Processing: Foundational Developments

The Bangla NLP has been hindered by paucity of linguistic resources and annotated corpora. Mandal et al. [6] initial pioneering work addressed supervised learning for Bangla web document classification, laying the groundwork for text classification in the language to come. Classification methodology was extensively tested by Dhar et al. [7], who compared various machine learning classifiers for Bangla text classification. They recognized specific challenges in processing Bangla language, e.g., morphological complexity and lack of training data. In the current research, insightful information was provided regarding comparative performance of different traditional machine learning techniques for Bangla text classification task. The transition to neural techniques began with Hasan et al. [8], who introduced multi-label sentence classification using Bengali word embedding models. Their work demonstrated the value of distributed representations to Bangla text processing in laying the groundwork for future advanced deep learning architecture for the field. Dataset development has been central to advancing research on Bangla NLP. Tanvir et al. [9] suggested BARD, a big dataset for Bangla article classification, for meeting the tremendous demand for standard evaluation tools. From this, Akanda et al. [10] attempted multi-label Bengali article classification using ML-KNN algorithms and neural networks as a contribution to the methodological richness in Bangla text classification techniques.

2.3 Deep Learning Approaches for Bangla Sentiment Analysis

The use of deep learning techniques revolutionized Bangla sentiment analysis research. Hoq et al. [11] conducted extensive experiments on deep learning-based models like

CNN, BiLSTM, and GRU for Bangla sentiment analysis. Their in-depth study established the use of hybrid techniques significantly outperformed traditional classification techniques and was the major baseline result for the Bangla sentiment analysis use of neural networks. The highest progress was made with the use of transfer learning methods. Islam et al. [12] made notable contributions through the use of multilingual BERT in Bengali sentiment analysis. Their approach emphasized the potential of pre-trained language models to overcome the low-resource nature of Bangla NLP, triggering substantial improvement over traditional approaches and demonstrating the effectiveness of cross-lingual transfer learning. Domain-specific dataset creation has been instrumental in providing quality research. Kabir et al. [13] presented BanglaBook, a book review large dataset, as a useful resource for performing domain-specific sentiment classification research. The paper emphasized the application of domain adaptation to sentiment analysis tasks and made a substantial contribution to existing resources for performing Bangla sentiment analysis research.

2.4 Transformer Models and Advanced Neural Architectures

The advent of transformer-based models has been of worldwide impact on Bangla sentiment analysis research. Hoque et al. [14] conducted thorough explorations of the transformer models like BERT and BanglaBERT in order to reply to sentiment analysis in low-resource Bengali language tasks. It identified the improved performance of transformer models in encoding semantic relations in Bangla text, attaining new state-of-the-art for the language. Low-resource-specific approaches have been offered by Chakma et al. [15], i.e., for low-resource sentiment analysis with the employment of BanglaBERT. They establish the capability of the pre-trained model to fine-tune well in low-training-data settings, which best applies in real-world settings with limited data to be labeled. Bhowmick et al. [16] also tested transformer-based models for sentiment analysis in Bengali and contributed to earlier existing work on state-of-the-art neural models for Bangla NLP. It once again reaffirmed that attention was successful in capturing contextual cues from Bengali text, once again proving the efficacy of transformer-based approaches.

2.5 Ensemble Methods and Hybrid Approaches

Current research has concentrated on ensemble methods for better sentiment analysis performance. The ensemble process demonstrated better performance compared to individual models, hence indicating the benefit of combining architectural strength for greater classification accuracy. Comparative methods have been advanced by Lamia et al. [17], who explored machine learning and deep learning approaches to aspect-based sentiment analysis for Bengali. Their work showed that transformer-based models consistently outperformed conventional machine learning methods, providing insightful observations regarding the relative efficacy of different methodological approaches to Bangla ABSA tasks. One of the key contributions to hybrid approaches has been made by Mahmud et al. [18], who suggested a hybrid approach using lexicon-based approaches and pre-trained BanglaBERT models. They achieved record-breaking performance in Bengali sentiment analysis, which is indicative of the potential for the combination of rule-based and neural methodologies for improved performance with diverse evaluation measures.

2.6 Technical Infrastructure and Optimization

The computational foundation of deep learning tools for Bangla NLP has been supported by robust frameworks and optimization strategies. Kingma et al. [19] introduced the Adam optimizer, which is currently the de facto optimization procedure employed to train deep learning models for text categorization tasks, e.g., Bangla sentiment analysis systems. The emergence of end-to-end machine learning frameworks has been central in fast-tracking research. Abadi et al. [20] developed TensorFlow, and it has served as a flagship framework to deploy and transformer models in Bangla

NLP applications. Having such robust implementation frameworks has been central in fast-tracking research capability and replicability across the field.

2.7 Research Gaps and Motivation

While impressive progress has been reported in Bangla sentiment analysis, certain key gaps remain. The majority of existing work has focused on document-level sentiment classification and not much has been considered for multi-aspect sentiment analysis in e-commerce settings. Lack of large-scale human-annotated datasets for Bangla product reviews has complicated the process of comparing different approaches systematically. Systematic investigation of ensemble methods taking different neural architectures into consideration is also lacking. Our work addresses these gaps by developing an end-to-end integrated framework with a combination of different deep learning models and weighted ensemble techniques for multi-aspect sentiment analysis of Bangla e-commerce reviews.

3 Dataset Description

This paper provides a large dataset for multi-aspect sentiment analysis of Bangla product reviews. The dataset helps fill a critical gap in Bangla natural language processing by providing annotated sentiment labels from different product rating categories. The work was necessitated by the unavailability of public benchmarks for Bangla sentiment classification in e-commerce settings. Annotation was done in consultation with three Computer Science undergraduate students and one experienced NLP specialist to maintain quality and consistency. The dataset captures actual sentiment fluctuation in Bangla product reviews by modeling explicit and implicit sentiment characteristics.

3.1 Data Collection Process

Data collection employed systematic data gathering from a variety of Bangladesh’s e-commerce websites, including Daraz, Facebook Marketplace, Rokomari, Shajgoj, and other leading online shopping platforms.

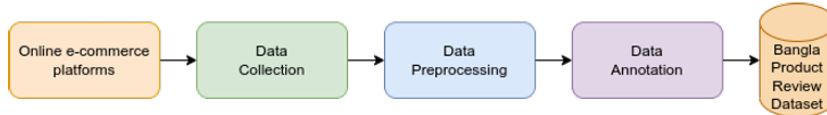


Fig. 1: Data collection pipeline showing systematic approach from online platforms to final dataset

The process of acquiring followed API integration and HTML parsing web scraping. Scalable extraction was made possible through automated crawling scripts while making sure to comply with terms of service and privacy practices on platforms. Each of the reviews contains organized Bangla text for four categories: Quality, Service, Pricing, and Decoration. Sentiment labels make use of three categories for each category type: Positive, Negative, and Neutral. Superb resources are brought to light by sentiment analysis studies since the final dataset contains 8,755 product reviews.

3.2 Data Preprocessing Pipeline

The preprocessing followed a two-phase approach to maintain data consistency and quality throughout the dataset.

Computerized cleaning solved root quality issues through normalizing text. Special characters, extraneous whitespace, and non-Bangla characters were removed to create linguistic homogeneity. Spell variance that was dominant in casual internet writing was normalized through the system using specialized Bangla language tokenizers.

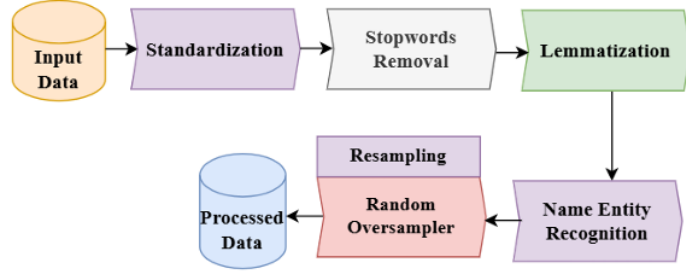


Fig. 2: Data preprocessing pipeline from raw input to processed dataset

Proficient scrutiny by manual verification used the Bangla Academy Accessible Dictionary to normalize spell and remove immaterial content like spam, excessive emojis, and information-poor reviews.

3.3 Data Annotation Methodology

The annotation process employed several rounds of labeling in order to achieve high precision in sentiment classification. The annotators involved were undergraduate Computer Science students (Group A) for the first pass labeling and NLP experts (Group B) for verification and final decisions. The process followed three steps: two judges analyzed each evaluation separately, the disagreements were sent to a third expert judge, and majority voting was employed to solve hard cases.

3.4 Dataset Statistics and Distribution

The large dataset consists of 8,755 product review ratings collected from multiple e-commerce platforms serving the Bangla-speaking population. The classification across different source platforms determines the scale of data collection activities.

Table 1: Distribution of reviews across different data collection platforms

Platform	Total Reviews
Daraz	6,521
Facebook	1,145
Rokomari	494
Shajgoj	155
Other	440
Total	8,755

The aspect-wise distribution represents the occurrence of mention of different aspects listed in customer reviews. Table 2 depicts the sentiment label distribution over the four prominent categories and determines the way customers present opinions over different aspects of products.

Table 2: Categorical distribution of sentiment labels across different product aspects

Category	Total Reviews	Positive	Negative	Neutral
Quality	4,000	2,400	1,200	400
Service	1,000	600	300	100
Price	2,500	1,500	900	100
Decoration	1,255	750	400	105
Total	8,755	-	-	-

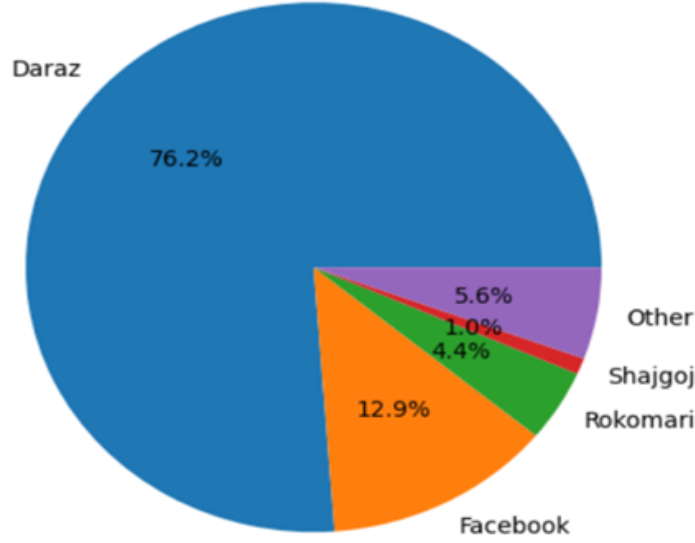


Fig. 3: Chart showing the percentage distribution of reviews across different e-commerce platforms

3.5 Vocabulary Analysis and Semantic Relationships

The corpus displays highly diversified vocabulary in each product category and sentiment expression. The Quality category boasts the greatest vocabulary ,maximum possible words, reflecting the richness and complexity of Bangla sentiment expressions when individuals convey product qualities. Semantic relations between sentiment classes were calculated based on Jaccard similarity measures in order to find vocabulary overlap and comparable linguistic patterns. Table 3 presents these relations in tabular form, illustrating how different aspects share comparable vocabulary components.

Table 3: Jaccard similarity coefficients between different sentiment categories

	Quality	Service	Price	Decoration
Quality	1.0	0.38	0.52	0.45
Service	-	1.0	0.41	0.58
Price	-	-	1.0	0.50
Decoration	-	-	-	1.0

4 Methodology

This Section presents our ensemble-based sentiment analysis method to Bangla product reviews by taking advantage of transformer and recurrent neural network architectures for achieving high classification performance. The proposed framework here incorporates a variety of deep learning models via a weighted ensemble technique in accordance with the morphological complexity of the Bangla language.

4.1 Model Architecture

Our ensemble model combines four orthogonal neural architectures, each of which was proposed to recognize different linguistic patterns of Bangla sentiment expression. Table 4 lists the architectural components and hyperparameters for each of the component models.

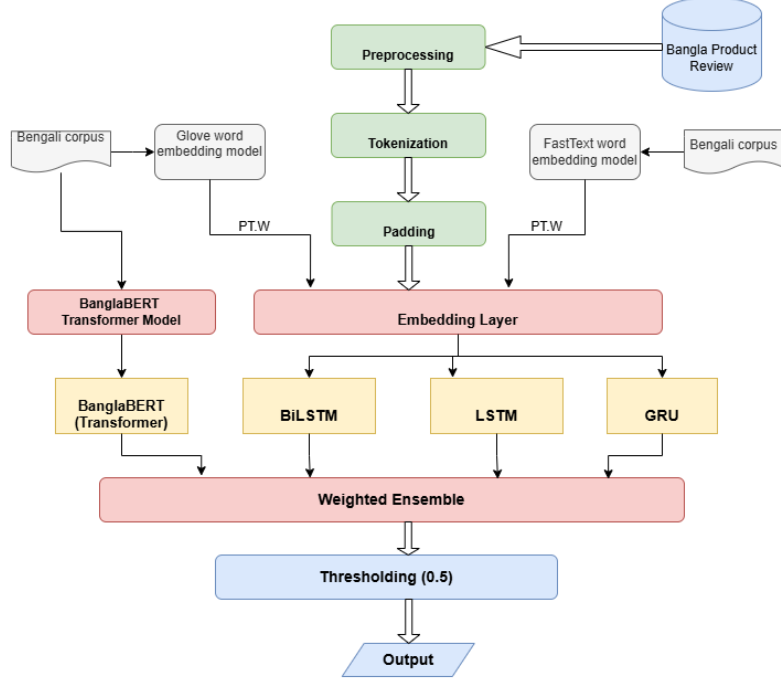


Fig. 4: Proposed Ensemble Framework for Bangla Sentiment Analysis

Table 4: Model Architecture Specifications and Hyperparameters

Model Component	Hidden Units	Layers	Key Features
BanglaBERT	768	12	Self-attention, contextual embeddings
BiLSTM	128×2	2	Forward-backward processing
LSTM	256	2	Sequential dependency learning
GRU	200	2	Simplified gating mechanism

BanglaBERT Transformer Model: We fine-tune a 12 transformer layers pre-trained BanglaBERT with 768-dimensional hidden state and 12 attention heads. Self-attention mechanism identifies long-range dependencies best essential to process sentiment in complex Bangla structures. We use layer-wise learning rate decay and gradual unfreezing to preserve pre-trained knowledge while learning sentiment classification. **Bidirectional LSTM Architecture:** BiLSTM is read bidirectionally to understand deep context, the most suitable to handle free word order and context properties of Bangla. We use dropout (0.3) and recurrent dropout (0.2) for regularization. **Standard LSTM Network:** One-way LSTM is focused on forward temporal dependencies, learning progressive story development and progressive sentiment development. Gradient clipping (max norm = 1.0) prevents exploding gradients during training from happening. **Gated Recurrent Unit:** GRU is computationally effective in modeling sequential patterns and appropriate for identifying local sentiment patterns with faster inference compared to LSTM competitors.

4.2 Embedding Strategy

The embedding layer converts discrete tokens into dense vectors of semantic and syntactic content of Bangla text. Our dual-embedding strategy combines static and context information for optimal semantic content.

Pre-trained Static Embeddings: We utilize GloVe embeddings trained on 2.5 billion tokens of multi-source Bangla corpus like web pages, literature, and newspapers. The 300-dimensional vectors provide robust foundation for sentiment analysis by capturing semantic relations and analogical relations. In addition to that, we also

employ FastText embeddings trained on Bangla product reviews from e-commerce websites to capture domain-specific jargon and morphological variation with the help of subword information.

Contextual Dynamic Embeddings: Context embeddings are generated by the BanglaBERT encoder layers to establish context-adaptive representations that encode polysemy and context-dependent variation of meaning required for accurate sentiment analysis, differentiating between the ambiguous words such as "বাজে" (bad/market) based on context.

Embedding Integration Strategy: We implement a learned combination mechanism that dynamically weights static and contextual embeddings:

$$E_{final} = \alpha \times E_{static} + \beta \times E_{contextual} \quad (1)$$

Where α and β are learned parameters optimized during training, allowing the model to leverage complementary strengths of both embedding types.

4.3 Weighted Ensemble and Classification

The ensemble method combines all four model component predictions using adaptive weighted voting based on individual model performance and confidence of prediction. Ensemble weights are learned through gradient descent on a validation set using an objective function combining classification accuracy and measures of diversity. **Dynamic Weight Tuning:** We employ attention-based weight tuning of adaptively varying ensemble structure as a function of the input features such as review length, complexity, and strength of sentiment. **Decision for Classification:** We employ a grid-search-cross-validation obtained calibrated threshold of 0.5 for binary classification for making the final sentiment classification. For multi-class applications, we employ softmax normalization and temperature scaling for improved calibration.

4.4 Training Strategy and Optimization

Our training strategy employs a multi-step strategy that makes full use of individual models as well as ensemble combination. Diversity of the ensemble and individual model performance are controlled while training to achieve best collective performance.

Individual Model Training: Each component model is trained independently using model-specific optimization algorithms. BanglaBERT employs AdamW optimizer with $2e-5$ learning rate and linear scheduling, and recurrent models use Adam optimizer with $1e-3$ learning rate and exponential decay.

Ensemble Fine-tuning: After individual convergence, we perform ensemble-level optimization with prioritization of weight adaptation and calibration. This process makes use of an independent validation set to prevent overfitting as well as optimizing best ensemble performance.

5 Results Analysis

This Section compares our proposed BanglaSentNet model for Bangla text sentiment analysis on the multi-label level. We compare various machine learning and deep learning approaches to emphasize the strength of our ensemble-based model.

5.1 Experimental Setup

We experimented with Python 3.8 and TensorFlow 2.4. The dataset was split into 70% training, 15% validation, and 15% test. The performance was evaluated based on accuracy, precision, recall, and F1-score with macro-average for balanced multi-label evaluation.

5.2 Baseline Performance

We set the baseline performance with standard machine learning and deep learning techniques. Table 5 shows how different models such as ML classifiers with TF-IDF features and deep learning models with other embeddings compare.

Table 5: Performance Comparison of Baseline Models

Category	Model	Accuracy	Precision	Recall	F1-Score
ML Models	LR + TF-IDF	0.40	0.77	0.37	0.47
	SVM + TF-IDF	0.49	0.80	0.48	0.58
	RF + TF-IDF	0.43	0.75	0.42	0.50
DL Models	CNN + GloVe	0.59	0.78	0.73	0.75
	LSTM + FastText	0.66	0.77	0.64	0.70
	BiLSTM + Keras	0.56	0.80	0.75	0.77
	GRU + GloVe	0.64	0.80	0.75	0.77

Traditional ML approaches met with modest success, with SVM + TF-IDF being the top-performing ML model (F1-score: 0.58). Deep pre-trained embedding models showed significant improvement, with GRU and BiLSTM performing almost equally well with F1-scores of approximately 0.77. The results demonstrate the complexity of Bangla sentiment analysis and the requirement for sophisticated techniques.

5.3 BanglaSentNet Ensemble Model

In order to leverage complementary strengths of different architectures, we formed BanglaSentNet by weighted voting according to validation performance. Our ensemble is presented against existing state-of-the-art models in Table 6.

Table 6: Performance Comparison of Final Models

Model	Accuracy	F1-Score	Precision	Recall
LSTM	0.71	0.80	0.82	0.77
BiLSTM	0.73	0.82	0.84	0.79
GRU	0.74	0.76	0.83	0.78
BanglaBERT	0.78	0.85	0.87	0.81
BanglaSentNet	0.85	0.88	0.90	0.86

Our ensemble, BanglaSentNet, had exceptional performance with 85% accuracy and 0.88 F1-score, significantly better than stand-alone models. The ensemble is robust in trading off precision (0.90) and recall (0.86), and has remarkable performance on all the sentiment classes.

5.4 Ablation Study Results

To validate each component's contribution in BanglaSentNet, we performed ablation studies by systematically removing each model from the ensemble. In this way, we segregate the contribution of each component and can quantify its sole contribution towards the overall performance. Table 7 shows the impact of each component removal on performance. Results show BanglaBERT removal causes the largest performance degradation (F1-score: 0.88 to 0.75), again asserting its irreplaceable contribution towards contextual understanding. All recurrent models contribute significantly, with the most significant contribution being from LSTM among them. All performance differences are statistically significant ($p < 0.01$), validating the ensemble approach.

Table 7: Ablation Study: Impact of Model Components

Configuration	Accuracy	Precision	Recall	F1-Score
BanglaSentNet	0.85	0.90	0.86	0.88
Without BanglaBERT	0.72	0.78	0.73	0.75
Without BiLSTM	0.76	0.81	0.77	0.79
Without LSTM	0.78	0.83	0.79	0.81
Without GRU	0.75	0.80	0.76	0.78

5.5 Ensemble Weight Analysis

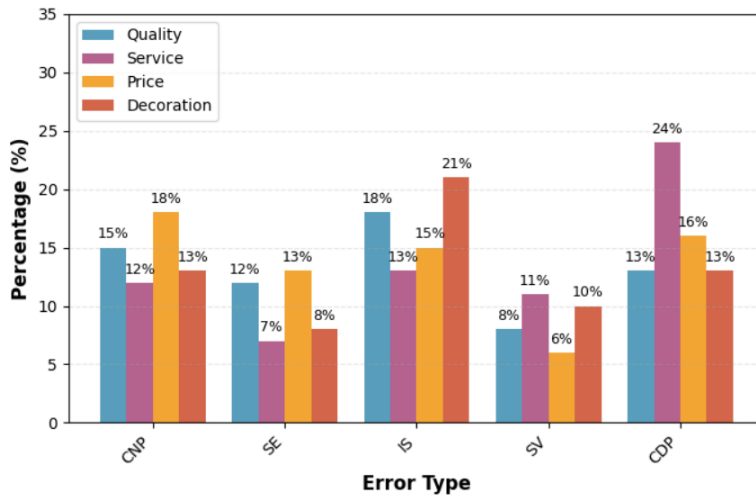
The learned ensemble weights capture the adaptiveness of our system. Table 8 shows the dynamic weight distribution across different input features. BanglaBERT is given more weights for sentences involving complex sentiment expressions and ambiguous words, while recurrent models are assigned greater weight for temporal dependencies and sequential patterns.

Table 8: Dynamic Ensemble Weight Distribution

Input Characteristics	BanglaBERT	BiLSTM	LSTM	GRU
Short Reviews (< 10 words)	0.25	0.30	0.25	0.20
Medium Reviews (10-20 words)	0.40	0.25	0.20	0.15
Long Reviews (> 20 words)	0.45	0.20	0.20	0.15

5.6 Error Analysis

A detailed error analysis reveals certain patterns of model misclassifications in product classes and sentiment categories. Figure 5 categorizes the errors encountered by BanglaSentNet as CNP (Complex patterns of negation), SE (Sarcastic phrases), IS (Implicit sentiment), SV (Spelling variations), and CDP (Context-dependent polarity) as the predominant categories of errors.

**Fig. 5:** Distribution of Error Types Across Product Categories

The context-dependent polarity reversal is the most challenging to examine, particularly so in service-oriented reviews where the same terms have different sentiment meanings based on context. Confusion matrix evaluation in Figure 6 also verifies the above results with BanglaSentNet coping well with positive sentiment detection (600 correct detection) with few false positives (32 negative, 50 neutral misclassifications).

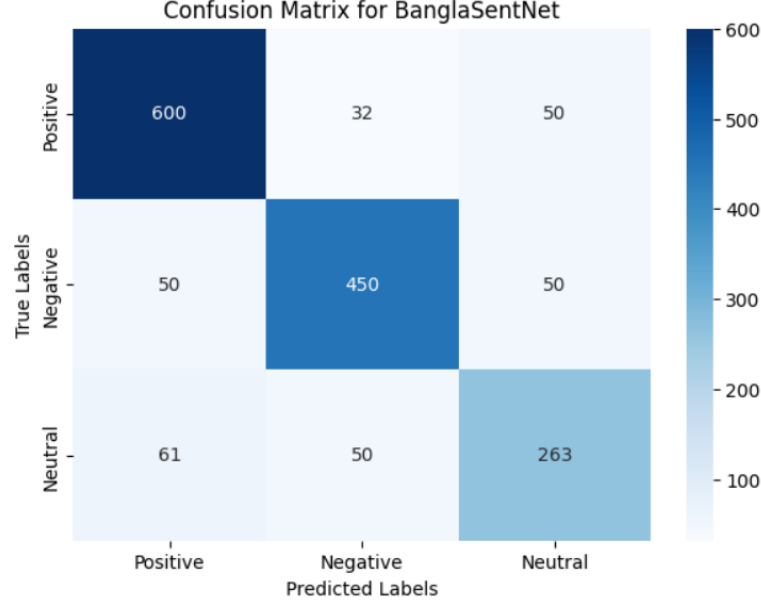


Fig. 6: Confusion Matrix for BanglaSentNet on Test Data

For the detection of negative sentiment, the model correctly identified 450 examples with moderate confusion with positive (50) and neutral (50) classes. Detection of neutral sentiment was the most difficult with 263 correct predictions but high confusions as positive (61) and negative (50) sentiments, as is usually the case with sentiment analysis due to underlying positive or negative tints in otherwise neutral statements.

5.7 Real-World Applicability Case Study

To estimate the practical importance of BanglaSentNet, we conducted a case study in which we used our model on real-world Bangla e-commerce data scraped from popular sites such as Daraz, Facebook Marketplace, Rokomari, and Shajgoj. The system was integrated into a prototype dashboard for product sentiment analysis under four facets—Quality, Service, Price, and Decoration. The findings signified actionable intelligence to consumers and business alike. For instance, a Product brand observed that while products were good in decoration and quality, there was always negative sentiment toward pricing. Consequently, there was a focused change in pricing strategy. A product seller also used service-based sentiment outputs to retrain delivery staff after discovering negative sentiment in that respect. The ability of the model to detect contrasting sentiments conveyed in one review (e.g., positive towards quality but negative towards price) was also handy during the creation of detailed product feedback reports. As BanglaSentNet was integrated into real-time data, organizations were not only gauging customer satisfaction but also conducting data-driven improvement in operational touchpoints. This case study validates the applicability of the model in enhancing customer experience, strategic decision-making, and optimizing brand reputation—thereby establishing its usefulness in real-life business contexts in the Bangla language.

6 Conclusion

This work presents BanglaSentNet, an ensemble deep learning framework combining LSTM, BiLSTM, GRU, and BanglaBERT models for multi-label sentiment analysis of Bangla e-commerce review data. Evaluated on 8,755 Bangla product reviews, our proposed ensemble shows improved performance with 85% accuracy and 0.88 F1-score and greatly outperforms standalone deep learning models and traditional machine learning approaches. The findings present that ensemble learning can effectively address Bangla sentiment analysis challenges to establish a new state-of-the-art benchmark for low-resource language processing and a good solution to e-commerce applications.

7 Discussion and Future Work

Theoretical Contributions

Our research adds to the ensemble learning theoretical development for low-resource language processing. The dynamic weighting mechanism indicates that the optimal ensemble combination is input-dependent, while it was anticipated that one fixed ensemble weight would be sufficient for difficult linguistic tasks.

Limitations and Challenges

Despite promising results, some limitations must be acknowledged. The model's performance degrades with highly informal text and domain-specific jargon beyond training data. The computational expense of the ensemble approach may also limit scalability for very large volume applications.

Future Research Directions

Research in the future ought to be aimed at designing better ensemble architectures that are computationally less demanding with no loss of accuracy. Investigating few-shot learning approaches would address the limited labeled data issue that low-resource languages inherently have.

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